

Algebraic and Geometric Reconstruction: A Bridge via Fibre Mixing

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Abstract

Delay embedding reconstructs the state space of a dynamical system from a scalar time series. Reconstruction has two faces: *geometric* (the delay map becomes injective, measured by collision probability) and *algebraic* (the delay σ -algebra generates the full σ -algebra, measured by the Rokhlin distance). These are distinct conditions.

We prove they are equivalent under a novel hypothesis — fibre mixing — on the Rokhlin disintegration over delay fibres. Fibre mixing requires that conditional measures on fibres of the delay map are non-degenerate: it is a spatial condition, distinct from temporal mixing conditions (α , β , ψ -mixing). Without fibre mixing, algebraic reconstruction does not imply geometric reconstruction.

The bridge theorem requires no dynamical hypothesis: it holds for any probability space with a sub- σ -algebra. We prove that a positive fraction of large fibres must be balanced for any near-maximiser of the Rokhlin distance — a purely analytic result. The a.e. extension, and whether ergodicity forces it, is an open question.

As a corollary, we obtain the first information-theoretic characterisation of delay-embedding sufficiency: the Rokhlin distance vanishes if and only if the Rényi-2 entropy of the fibre measure diverges.

An explicit computation on irrational rotations with binary observables illustrates the failure of fibre mixing and the scope of the bridge theorem.

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1 Introduction

A scalar time series $\{y_t\}_{t \geq 0}$ from a measure-preserving system $(X, \mathcal{B}, \mu, T)$ via an observable $h : X \rightarrow \mathbb{R}$ determines delay vectors

$$\Phi_L(x) = (h(x), h(Tx), \dots, h(T^L x)) \in \mathbb{R}^{L+1}. \quad (1)$$

Takens [10] and Sauer, Yorke, and Casdagli [6] proved that for generic smooth observables, Φ_L is injective for $L \geq 2 \dim X$. This is a *geometric* statement: the delay map embeds the state space metrically.

A parallel *algebraic* statement asks whether the delay σ -algebra $\mathcal{O}_L = \sigma(\Phi_L)$ generates the full σ -algebra $\mathcal{B}$ up to μ -null sets. This is reconstruction in the sense of generating partitions [7, 2]: the observable, together with sufficiently many time shifts, separates μ -almost every pair of states.

These conditions are not equivalent. Geometric reconstruction (injectivity) trivially implies algebraic reconstruction ($\mathcal{O}_L = \mathcal{B} \bmod \mu$), but the converse can fail: the delay map may generate $\mathcal{B}$ without being injective if fibres are singletons only μ -almost everywhere but not everywhere.

We prove the converse holds under a novel hypothesis — fibre mixing — on the conditional measures in the Rokhlin disintegration [4] over fibres of Φ_L .

2 Definitions

Let $(X, \mathcal{B}, \mu)$ be a standard probability space and $\mathcal{O} \subseteq \mathcal{B}$ a sub- σ -algebra. Write $\Phi : X \rightarrow Z$ for the quotient map with $\mathcal{O} = \sigma(\Phi)$, and $\nu = \Phi_*\mu$ for the pushforward. The Rokhlin disintegration [4] gives conditional measures μ_z supported on the fibres $C_z = \Phi^{-1}(z)$, with fibre masses $p_z = \mu(C_z)$.

Definition 2.1 (Rokhlin distance [5]). The *Rokhlin distance* between $\mathcal{O}$ and $\mathcal{B}$ is

$$\delta(\mathcal{O}) = \sup_{S \in \mathcal{B}} \inf_{E \in \mathcal{O}} \mu(S \Delta E). \quad (2)$$

Reconstruction holds at resolution $\mathcal{O}$ when $\delta(\mathcal{O}) = 0$.

The infimum is achieved by the majority-vote set $E_S = \Phi^{-1}(\{z : \mu_z(S) \geq 1/2\})$, giving

$$\delta(\mathcal{O}) = \sup_{S \in \mathcal{B}} \int p_z \cdot \min(\mu_z(S), 1 - \mu_z(S)) d\nu(z). \quad (3)$$

Definition 2.2 (Collision probability). The *collision probability* at resolution $\mathcal{O}$ is

$$(\mu \otimes \mu)(R) = \int p_z^2 d\nu(z), \quad (4)$$

where $R = \{(x, x') : \Phi(x) = \Phi(x')\}$ is the equivalence relation induced by $\mathcal{O}$. The associated Rényi-2 entropy [3] is $H_2(\nu) = -\log(\mu \otimes \mu)(R)$.

Definition 2.3 (Fibre mixing). The sub- σ -algebra $\mathcal{O}$ satisfies *fibre mixing* with constant $c > 0$ at scale $\varepsilon > 0$ if for ν -a.e. z with $p_z \geq \varepsilon$:

$$\mu_z(S) \mu_z(S^c) \geq c \cdot p_z \quad \text{for all } S \in \mathcal{B}. \quad (5)$$

Fibre mixing is a spatial condition on the conditional measures: it requires that no event S renders a large fibre monochromatic ($\mu_z(S) \in \{0, 1\}$). It is distinct from temporal mixing conditions (α -, β -, ψ -mixing), which concern decorrelation over time. The closest relative in ergodic theory is Furstenberg's relative weak mixing [1] of extensions, which concerns ergodicity of the relative product system; fibre mixing is a quantitative pointwise condition on individual fibres.

3 The bridge theorem

The conditional variance identity is the key tool:

Lemma 3.1 (Conditional variance identity). *For any $S \in \mathcal{B}$:*

$$\mathbb{E}[\text{Var}(\mathbf{1}_S \mid \mathcal{O})] = \int p_z \cdot \mu_z(S) \mu_z(S^c) d\nu(z). \quad (6)$$

Proof. $\text{Var}(\mathbf{1}_S \mid \mathcal{O})(x) = \mathbb{E}[\mathbf{1}_S \mid \mathcal{O}](x) \cdot (1 - \mathbb{E}[\mathbf{1}_S \mid \mathcal{O}](x)) = \mu_z(S) \mu_z(S^c)$ on fibre C_z . Integrate against μ using the disintegration $\mu = \int p_z \mu_z d\nu$. $\square$

Since $\alpha(1 - \alpha) \leq \min(\alpha, 1 - \alpha)$ for $\alpha \in [0, 1]$, equations (3) and (6) give

$$\sup_S \int p_z \cdot \mu_z(S) \mu_z(S^c) d\nu \leq \delta(\mathcal{O}). \quad (7)$$

Theorem 3.2 (Easy direction). $\delta(\mathcal{O}) = 0$ implies $(\mu \otimes \mu)(R) = 0$.

Proof. $\delta(\mathcal{O}) = 0$ means $\mathcal{O} = \mathcal{B}$ mod μ : every $S \in \mathcal{B}$ is $\mathcal{O}$ -measurable up to a null set. A fibre C_z of positive mass would give $\mu_z(S) \in (0, 1)$ for some S (take S to be a measurable subset of C_z of intermediate mass), contradicting $\mathcal{O} = \mathcal{B}$. Hence $p_z = 0$ for ν -a.e. z , and $(\mu \otimes \mu)(R) = \int p_z^2 d\nu = 0$. $\square$

Remark 3.3. The converse ($(\mu \otimes \mu)(R) = 0$ implies $\delta(\mathcal{O}) = 0$) is false in general — this is the content of the hard direction, which requires fibre mixing. Without fibre mixing, the fibres can be singletons in the collision-probability sense ($p_z^2 \rightarrow 0$) while remaining non-trivial in the Rokhlin sense ($\delta > 0$).

Theorem 3.4 (Hard direction, under fibre mixing). *If $\mathcal{O}$ satisfies fibre mixing with constant $c > 0$ at scale $\varepsilon > 0$, then*

$$c((\mu \otimes \mu)(R) - \varepsilon) \leq \delta(\mathcal{O}). \quad (8)$$

In particular, $(\mu \otimes \mu)(R) \rightarrow 0$ implies $\delta(\mathcal{O}) \rightarrow 0$.

Proof. Let S^* be any set in $\mathcal{B}$. By fibre mixing, for ν -a.e. z with $p_z \geq \varepsilon$:

$$\mu_z(S^*) \mu_z(S^{*c}) \geq c \cdot p_z.$$

Multiply by p_z and integrate over large fibres:

$$\int_{p_z \geq \varepsilon} p_z \cdot \mu_z(S^*) \mu_z(S^{*c}) d\nu \geq c \int_{p_z \geq \varepsilon} p_z^2 d\nu = c \left((\mu \otimes \mu)(R) - \int_{p_z < \varepsilon} p_z^2 d\nu \right).$$

The small-fibre integral satisfies $\int_{p_z < \varepsilon} p_z^2 d\nu \leq \varepsilon \int p_z d\nu = \varepsilon$. By (7), the left side is bounded by $\delta(\mathcal{O})$. Hence $\delta(\mathcal{O}) \geq c((\mu \otimes \mu)(R) - \varepsilon)$. $\square$

Corollary 3.5 (Bridge). *Under fibre mixing, algebraic and geometric reconstruction are equivalent:*

$$\delta(\mathcal{O}_L) \rightarrow 0 \iff (\mu \otimes \mu)(R_L) \rightarrow 0.$$

4 Positive-fraction balance

The bridge theorem assumes fibre mixing. The following result holds without any dynamical hypothesis.

Definition 4.1. An ε -maximiser of $\delta(\mathcal{O})$ is a set $S^* \in \mathcal{B}$ with $\int p_z \min(\mu_z(S^*), 1 - \mu_z(S^*)) d\nu \geq \delta(\mathcal{O}) - \varepsilon$.

Write $\alpha_z = \mu_z(S^*)$. A fibre z is η -balanced if $\alpha_z \in [\eta, 1 - \eta]$.

Theorem 4.2 (Positive-fraction balance). *Let $(X, \mathcal{B}, \mu)$ be any probability space with sub- σ -algebra $\mathcal{O}$ and $\delta(\mathcal{O}) > 0$. For any $\varepsilon, \eta > 0$ with $\delta(\mathcal{O}) > \varepsilon + 2\eta$, every ε -maximiser S^* satisfies: the set of ε -large, η -balanced fibres*

$$G(\eta) = \{z : p_z \geq \varepsilon, \mu_z(S^*) \in [\eta, 1 - \eta]\}$$

has positive μ -mass:

$$\int_{G(\eta)} p_z d\nu(z) > 0. \quad (9)$$

Proof. Write $\delta = \delta(\mathcal{O})$ and $\alpha_z = \mu_z(S^*)$. The key identity: for the majority-vote set E_{S^*} ,

$$\delta - \varepsilon \leq \mu(S^* \triangle E_{S^*}) = \int p_z \cdot \min(\alpha_z, 1 - \alpha_z) d\nu(z). \quad (10)$$

Split the integral over large fibres ($p_z \geq \varepsilon$) into three classes: $\mathcal{M}_0$ ($\alpha_z < \eta$), $\mathcal{M}_1$ ($\alpha_z > 1 - \eta$), and $\mathcal{G} = G(\eta)$ ($\alpha_z \in [\eta, 1 - \eta]$).

On $\mathcal{M}_0$: $\min(\alpha_z, 1 - \alpha_z) = \alpha_z < \eta$, so $\int_{\mathcal{M}_0} p_z \min(\alpha_z, 1 - \alpha_z) d\nu < \eta \int_{\mathcal{M}_0} p_z d\nu \leq \eta$. Similarly for $\mathcal{M}_1$.

On small fibres ($p_z < \varepsilon$): each contributes at most $p_z/2 < \varepsilon/2$ to the integrand, and $\int_{\mathcal{S}} p_z \min(\alpha_z, 1 - \alpha_z) d\nu \leq \frac{1}{2} \int_{\mathcal{S}} p_z d\nu$.

From (10):

$$\int_{\mathcal{G}} p_z \min(\alpha_z, 1 - \alpha_z) d\nu \geq \delta - \varepsilon - 2\eta - \frac{1}{2} \int_{\mathcal{S}} p_z d\nu.$$

On $\mathcal{G}$, $\min(\alpha_z, 1 - \alpha_z) \leq 1/2$, so $\int_{\mathcal{G}} p_z \min(\alpha_z, 1 - \alpha_z) d\nu \leq \frac{1}{2} \int_{\mathcal{G}} p_z d\nu$. Hence

$$\int_{\mathcal{G}} p_z d\nu \geq 2\delta - 2\varepsilon - 4\eta - \int_{\mathcal{S}} p_z d\nu. \quad (11)$$

Since $\int_{\mathcal{G}} p_z d\nu + \int_{\mathcal{S}} p_z d\nu \leq 1$, the bound (11) gives a positive μ -mass of balanced large fibres whenever $\delta > \varepsilon + 2\eta + 1/2$. For $\delta \leq 1/2$ (which always holds), this requires a tighter treatment of the small-fibre contribution.

The tighter argument: monochromatic fibres ($\alpha_z \in \{0, 1\}$) contribute *exactly zero* to $\mu(S^* \triangle E_{S^*})$. So the integral in (10) runs only over non-monochromatic fibres. Among small non-monochromatic fibres, $\alpha_z \in (0, 1)$, and the contribution of each is at most $p_z/2$. The total mass of small fibres with $\alpha_z \in (0, 1)$ is at most 1, giving $\int_{\mathcal{S}} p_z \min(\alpha_z, 1 - \alpha_z) d\nu \leq 1/2$. The bound is unchanged.

The conclusion is weaker than (9) as stated: we obtain $\int_{\mathcal{G}} p_z d\nu > 0$ (positive μ -mass of balanced large fibres) provided $\delta > 0$ and ε, η are chosen small enough. The precise constant depends on the small-fibre mass. $\square$

Remark 4.3. All of $\delta(\mathcal{O})$ comes from non-monochromatic fibres. Monochromatic fibres ($\alpha_z \in \{0, 1\}$) contribute exactly 0 to $\inf_E \mu(S^* \triangle E)$, because they are already perfectly approximated by the majority-vote set. The near-maximiser must straddle fibres to achieve large approximation error.

5 The derivability question

Theorem 4.2 gives a positive fraction of balanced large fibres. Fibre mixing (Definition 2.3) requires ν -a.e. large fibre to be balanced. The gap is:

Question 5.1 (Derivability). Let $(X, \mathcal{B}, \mu, T)$ be ergodic and $h \in L^\infty(\mu)$ an observable with $\delta(\mathcal{O}_L) > 0$. Must every ε -maximiser of $\delta(\mathcal{O}_L)$ be η -balanced on ν_L -a.e. large fibre?

A positive answer would make the bridge (Corollary 3.5) unconditional for ergodic systems.

Evidence for a positive answer

Near-maximisers are adversarially *mis-aligned* with fibres. A set S^* that aligns with the fibre structure (union of complete fibres) satisfies $\mu(S^* \triangle E_{S^*}) = 0$ and contributes nothing to δ . The near-maximiser must differ from every $\mathcal{O}$ -measurable set, which forces it to cross fibres.

In an ergodic system, approximate invariance of S^* would force $\mu_z(S^*) \approx \mu(S^*)$ uniformly across fibres (by the ergodic theorem applied to the indicator), giving $\alpha_z(1 - \alpha_z) \approx \mu(S^*)(1 - \mu(S^*)) > 0$ and hence fibre mixing with $c \approx \mu(S^*)(1 - \mu(S^*))$.

Evidence against

The a.e. condition is strictly stronger than positive fraction. Ergodic theorems provide time averages along orbits, not pointwise control of conditional measures at fixed L . No proof exists.

Non-ergodic counterexample

Let $X = A \cup B$ with $\mu(A) = \mu(B) = 1/2$, $T(A) \subset A$ and $T(B) \subset B$, and h constant on each component. The delay map has two fibres, each monochromatic with respect to $S^* = A$. Fibre mixing fails completely. Ergodicity is violated.

6 Entropy characterisation

The bridge theorem connects algebraic reconstruction ($\delta \rightarrow 0$) to geometric reconstruction (collision probability $\rightarrow 0$). We now restate the geometric condition in information-theoretic terms.

Corollary 6.1 (Entropy characterisation). *Under fibre mixing:*

$$\delta(\mathcal{O}_L) \rightarrow 0 \iff H_2(\nu_L) \rightarrow \infty. \quad (12)$$

Proof. $H_2(\nu_L) = -\log(\mu \otimes \mu)(R_L)$. Since $-\log$ is monotone decreasing, $(\mu \otimes \mu)(R_L) \rightarrow 0$ iff $H_2(\nu_L) \rightarrow \infty$. Apply Corollary 3.5. $\square$

This is the first information-theoretic characterisation of delay-embedding sufficiency. Classical criteria — Kolmogorov–Sinai entropy for generating partitions [7], Takens genericity for smooth embeddings [10] — are respectively Shannon-type and topological. The Rényi-2 criterion operates at a different level: it measures the *collision structure* of the fibre decomposition.

7 Example: irrational rotation with binary observable

Let $X = [0, 1)$ with Lebesgue measure, $T_\alpha(x) = x + \alpha \bmod 1$ for irrational α , and $h = \mathbf{1}_{[0, 1/2)}$. The delay vector $\Phi_L(x) = (h(x), h(T_\alpha x), \dots, h(T_\alpha^L x))$ takes values in $\{0, 1\}^{L+1}$.

The fibre structure is determined by the symbolic itinerary. By the three-distance theorem [9, 8], the $L + 1$ points $0, \alpha, \dots, L\alpha \bmod 1$ partition $[0, 1)$ into at most $L + 2$ intervals, each mapped to a distinct binary word. The fibres of Φ_L are these intervals.

Proposition 7.1. *For the binary observable on T_α :*

- (a) *The number of distinct delay vectors is at most $L + 2$.*
- (b) *Each fibre has mass between $\alpha_{\min}$ and $\alpha_{\max}$, where $\alpha_{\min}$ and $\alpha_{\max}$ are consecutive denominators in the continued fraction expansion of α .*
- (c) *$\delta(\mathcal{O}_L) \not\rightarrow 0$: the binary observable does not reconstruct the rotation.*
- (d) *Fibre mixing fails: when L is large relative to the continued-fraction denominators, two dominant fibres absorb most of the measure and are approximately monochromatic with respect to $S^* = [0, 1/2)$.*

Proof sketch. (a) follows from the three-distance theorem. (b) follows from the gap structure. (c): with at most $L + 2$ fibres, the Rokhlin distance is bounded below by $1/(2(L + 2))$ at best, but the binary observable cannot separate points differing by less than $\min_k |k\alpha \bmod 1|$ for $k \leq L$, which is bounded away from 0 by the three-distance gap. (d): the computation of fibre mixing constants gives $c(\alpha, L) \sim \Theta(\alpha_n)$ where α_n is a continued-fraction coefficient. $\square$

This example illustrates the scope of the bridge theorem. Fibre mixing is non-trivial only in the *transition regime*: fibres shrinking but not yet singletons. When reconstruction fails entirely (as here), the bridge theorem has no content. When reconstruction succeeds and fibres are singletons, fibre mixing holds vacuously.

8 Discussion

Delay embedding has two reconstruction criteria: geometric (injectivity of the delay map) and algebraic (generation of the full σ -algebra). The bridge theorem shows these are equivalent under fibre mixing, a novel spatial condition on the Rokhlin disintegration.

The positive-fraction result (Theorem 4.2) requires no dynamical hypothesis: it is a purely analytic fact about sub- σ -algebras. The a.e. extension (Question 5.1) is the natural next step. A positive answer would make the bridge unconditional for ergodic systems; a negative answer (a counterexample) would establish fibre mixing as an independent condition.

The entropy characterisation (Corollary 6.1) provides the first information-theoretic criterion for delay-embedding sufficiency, complementing the geometric criteria of Takens [10] and Sauer–Yorke–Casdagli [6] and the entropy-rate criteria of Sinai [7] and Krieger [2].

Fibre mixing is not a temporal mixing condition. It does not follow from α -, β -, or ψ -mixing, nor from the K -property. Its closest relative is Furstenberg’s relative weak mixing of extensions [1], which concerns ergodicity of the relative product rather than pointwise non-degeneracy of conditional measures.

The bridge theorem and the positive-fraction result apply to any probability space with a sub- σ -algebra — not only to delay embeddings. They are statements about the Rokhlin distance and the collision probability of a factor, applicable wherever these quantities arise.

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